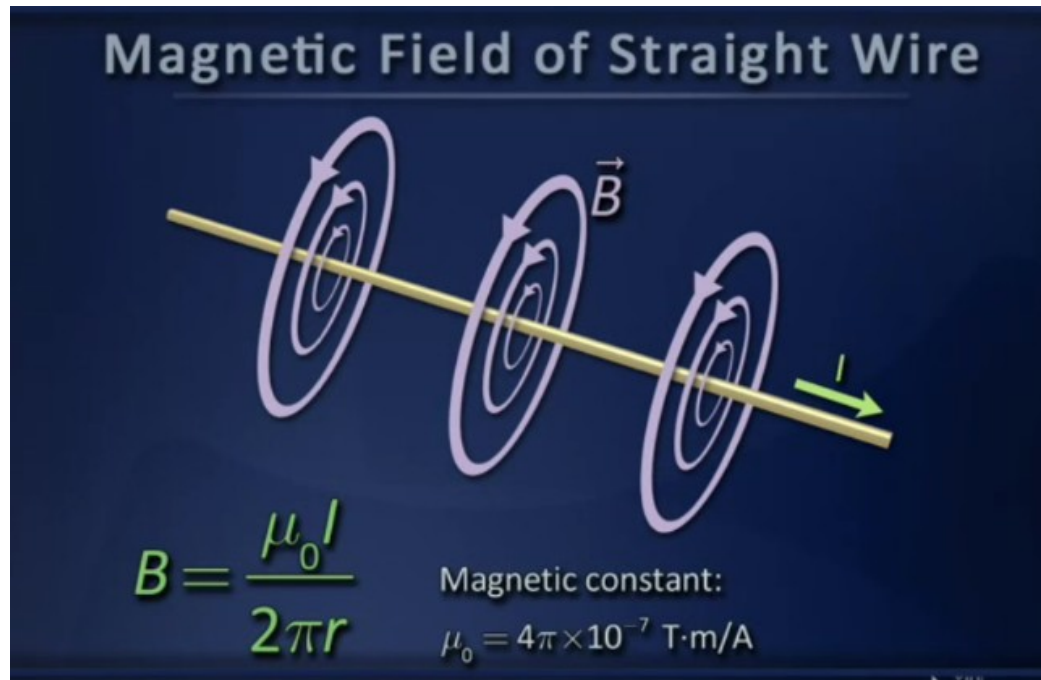
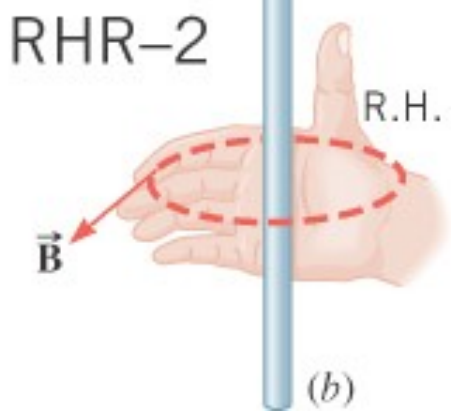
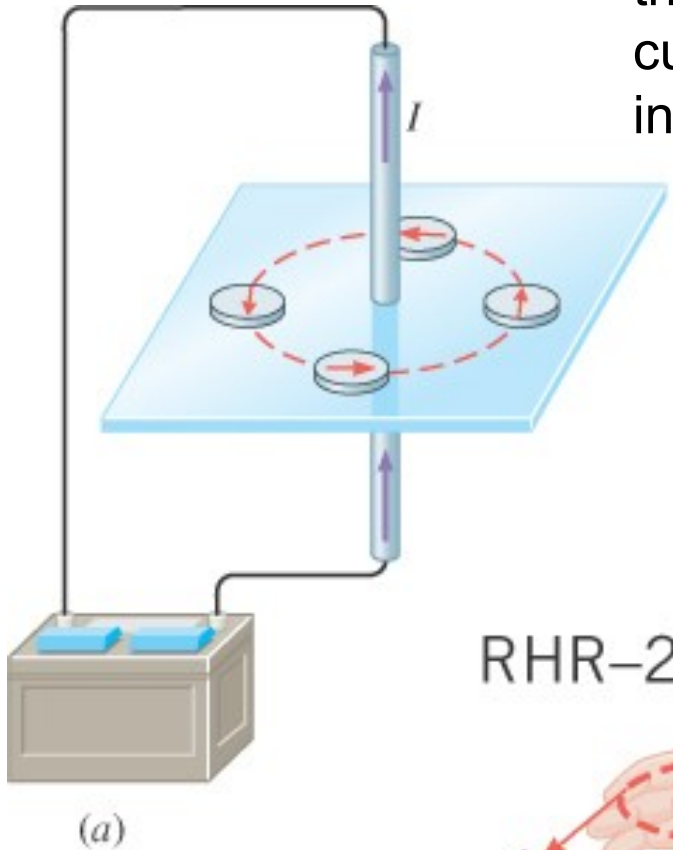


Source of magnetism



21.7 Magnetic Fields Produced by Currents

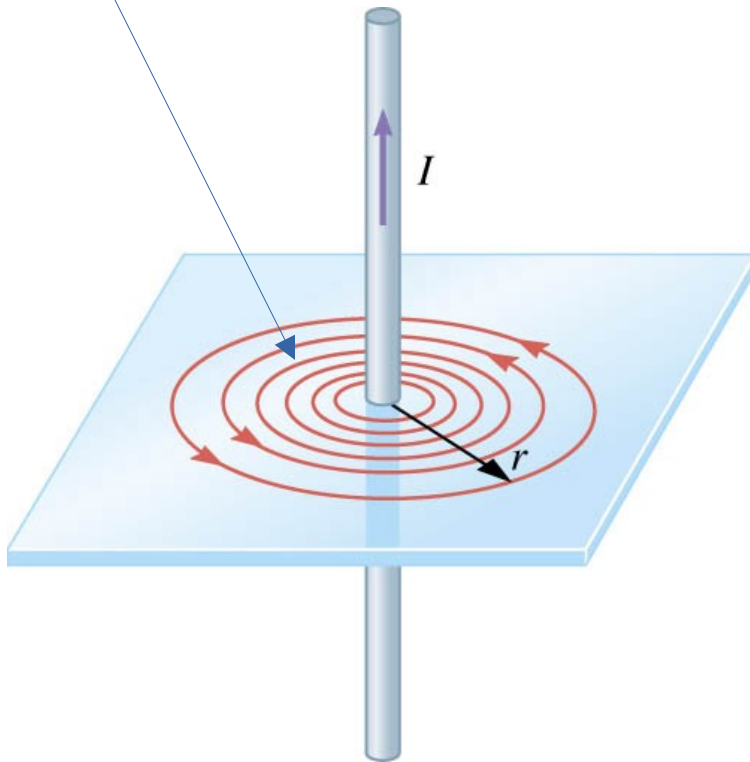
Right-Hand Rule No. 2. Curl the fingers of the right hand into the shape of a half-circle. Point the thumb in the direction of the conventional current, and the tips of the fingers will point in the direction of the magnetic field.



21.7 Magnetic Fields Produced by Currents

A LONG, STRAIGHT WIRE

See how the lines are more packed close
To wire $\rightarrow$ larger density $\rightarrow$ larger B



$$B = \frac{\mu_o I}{2\pi r} = \frac{2 \cdot 10^{-7} I}{r}$$

$$\mu_o = 4\pi \times 10^{-7} \text{ T} \cdot \text{m} / \text{A}$$

permeability of
free space

A magnetometer placed 1 centimeter from a wire indicates that the magnetic field at that position is 0.30 milliteslas. How much is the current in the wire?

Find the direction and magnitude of the magnetic field 12cm above a wire Carrying a current of 6A northward.

How much current is needed to produce a field of 0.25mTs at a distance Of 20cm from a heavy wire that is carrying the current?
(a lot)

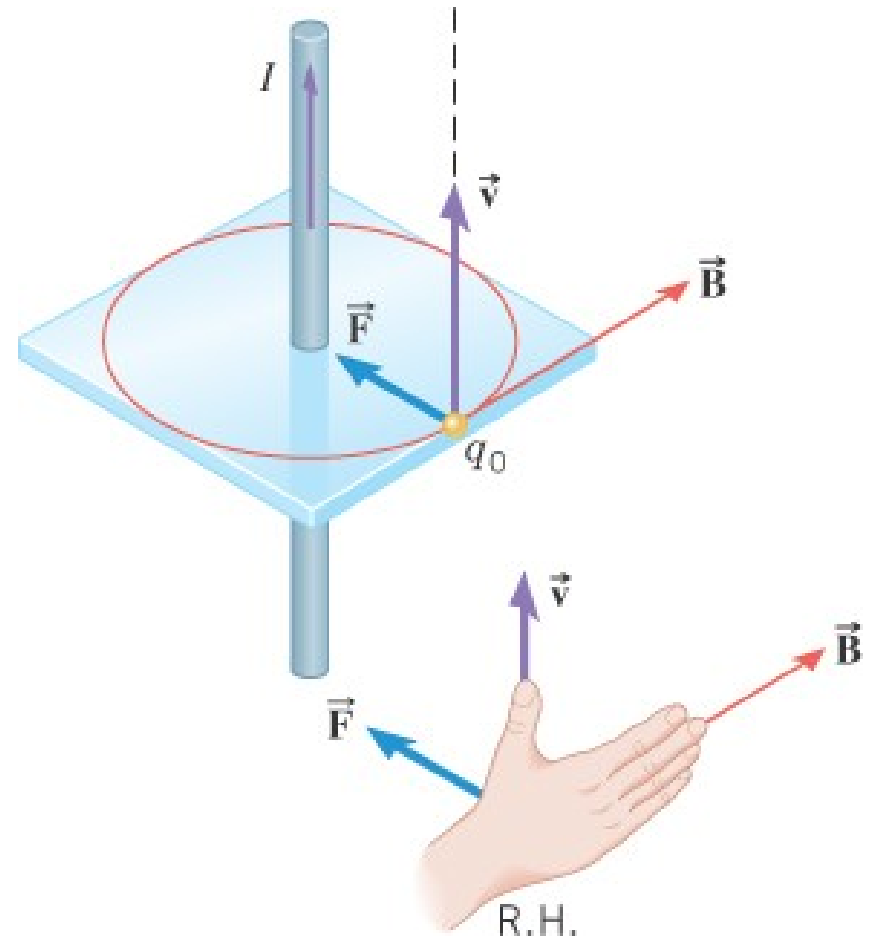
Find the direction and magnitude of the magnetic field 5cm above a wire carrying 20 A to to the west.

21.7 Magnetic Fields Produced by Currents

Example 7 A Current Exerts a Magnetic Force on a Moving Charge

The long straight wire carries a current of 3.0 A. A particle has a charge of $+6.5 \times 10^{-6}$ C and is moving parallel to the wire at a distance of 0.050 m. The speed of the particle is 280 m/s.

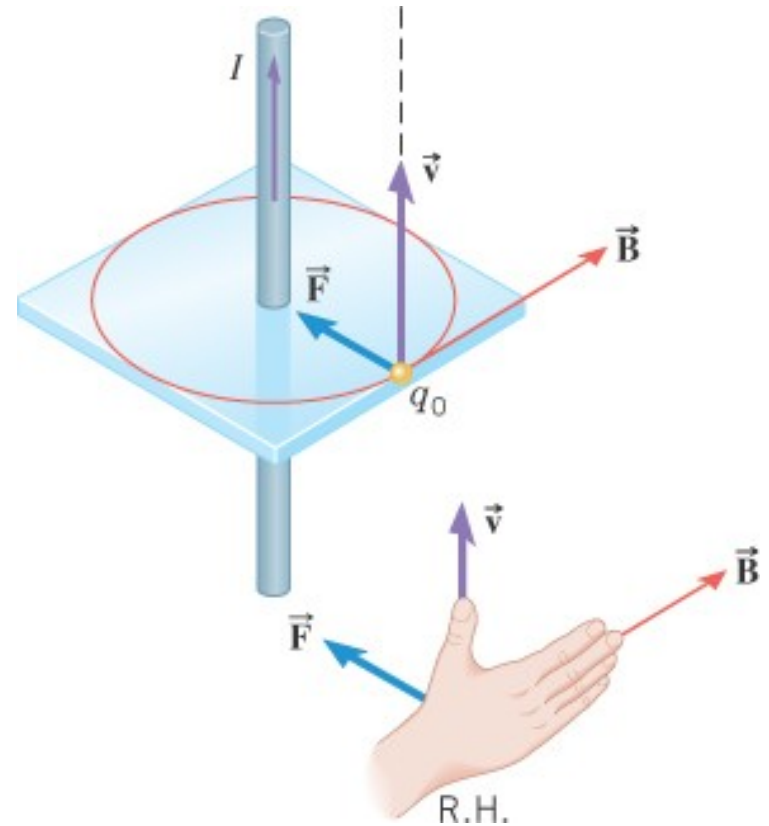
Determine the magnitude and direction of the magnetic force on the particle.



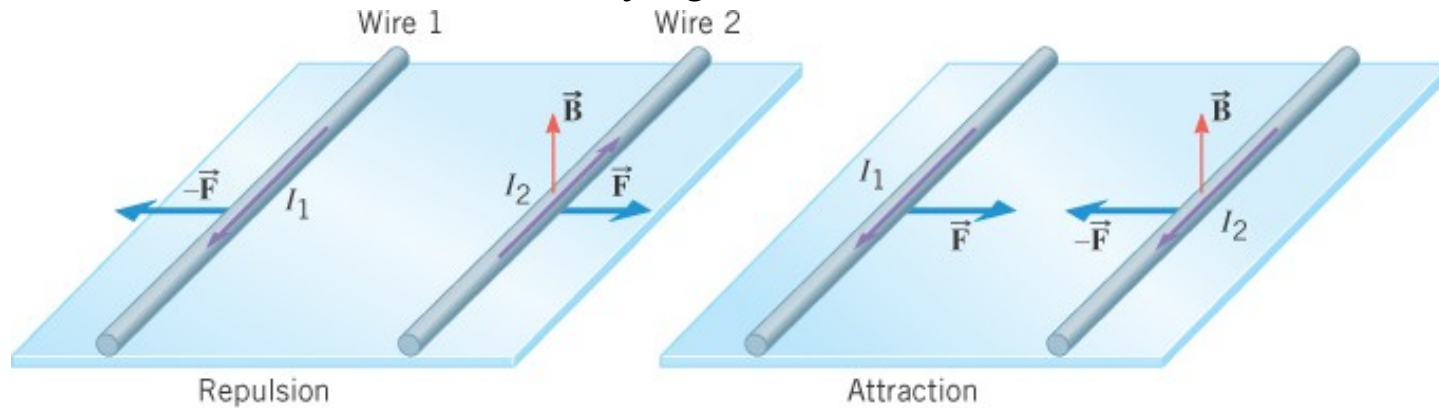
21.7 Magnetic Fields Produced by Currents

$$F = qvB \sin \theta = qv \left(\frac{\mu_o I}{2\pi r} \right) \sin \theta$$

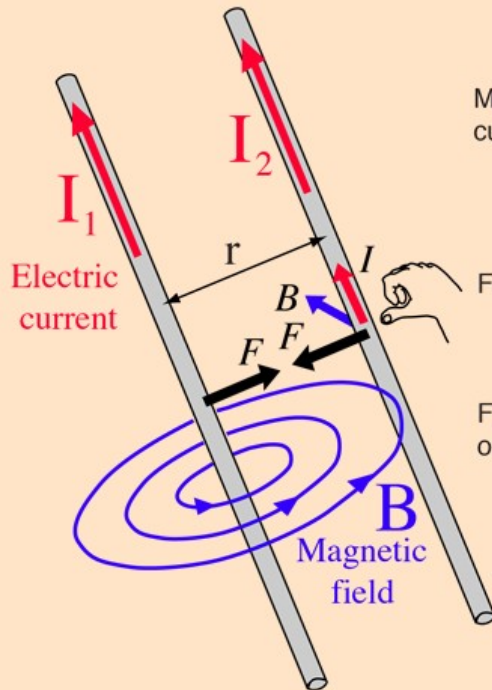
$B = \frac{\mu_o I}{2\pi r}$



Current carrying wires can exert forces on each other.



Magnetic Force Between Wires



Magnetic field at wire 2 from current in wire 1:

$$B = \frac{\mu_0 I_1}{2\pi r}$$

Force on a length ΔL of wire 2:

$$F = I_2 \Delta L B$$

Force per unit length in terms of the currents:

$$\frac{F}{\Delta L} = \frac{\mu_0 I_1 I_2}{2\pi r}$$

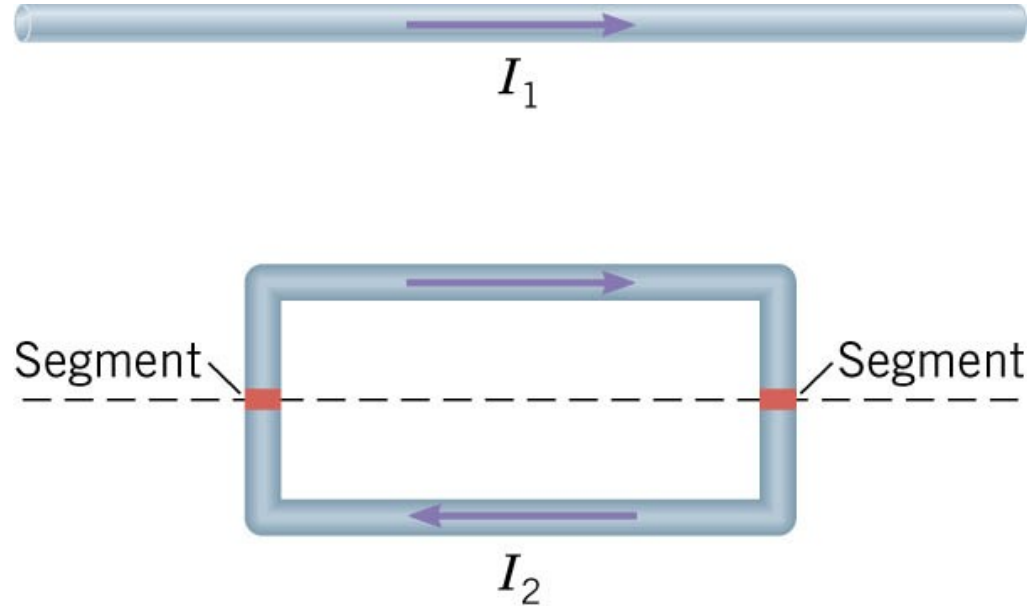
Show:

$$F/L = (I_1 I_2) (2 \cdot 10^{-7}) / r$$

21.7 Magnetic Fields Produced by Currents

Conceptual Example 9 The Net Force That a Current-Carrying Wire Exerts on a Current Carrying Coil

Is the coil attracted to, or repelled by the wire?

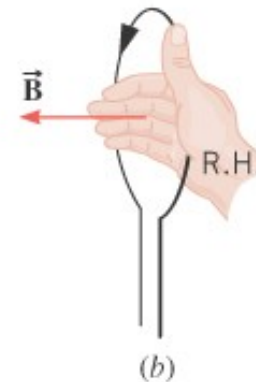
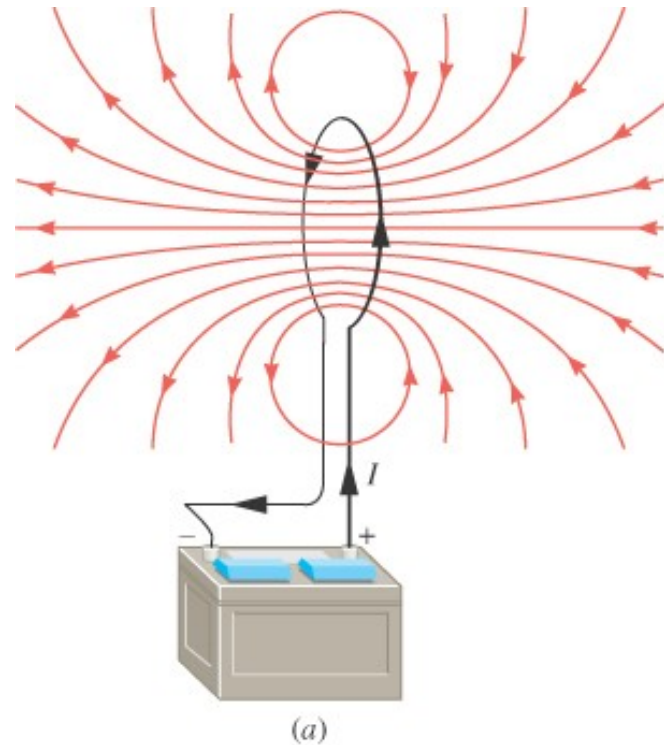


21.7 Magnetic Fields Produced by Currents

A LOOP OF WIRE

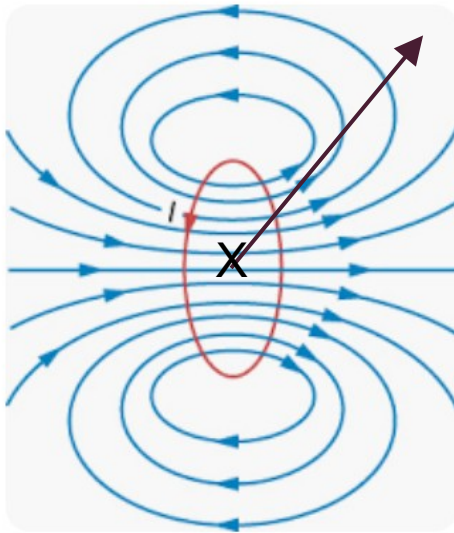
$$B = \frac{\mu_o I}{2R}$$

center of circular loop

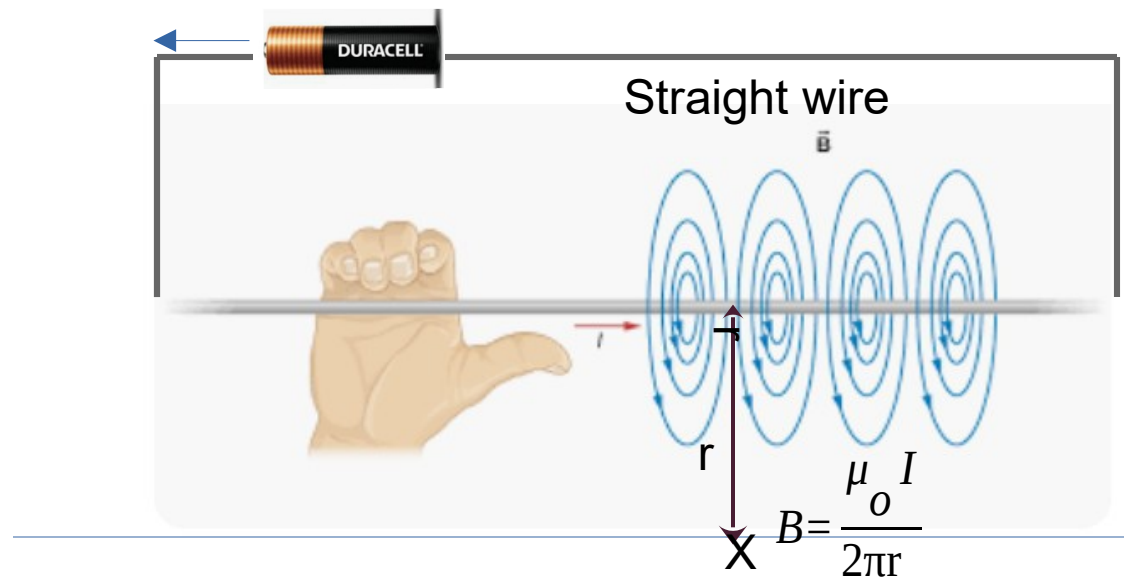
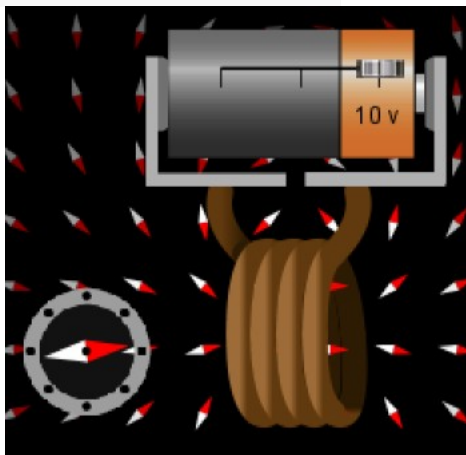


loop/few loops

$$B = \frac{\mu_0 I}{2R}$$



Where is the North?

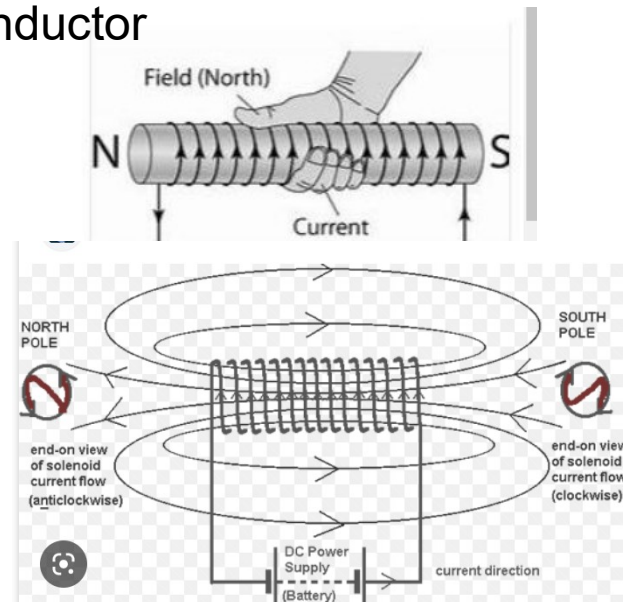


solenoid/coil/inductor



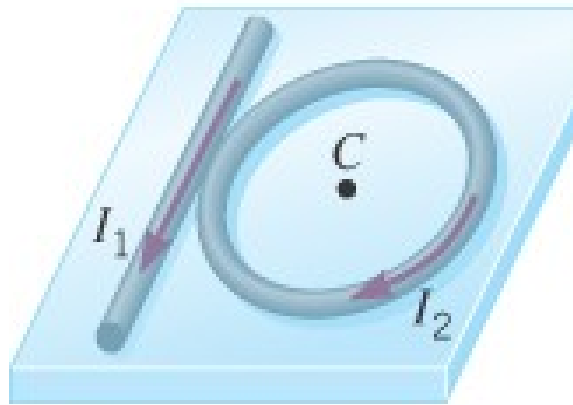
$$B = \mu_0 nI$$

B Inside is uniform
Outside is almost 0

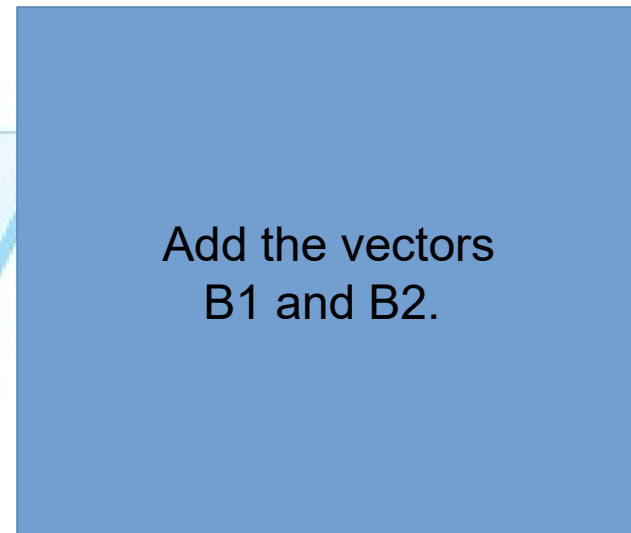


Example 10 Finding the Net Magnetic Field

A long straight wire carries a current of 8.0 A and a circular loop of wire carries a current of 2.0 A and has a radius of 0.030 m. Find the magnitude and direction of the magnetic field at the center of the loop.



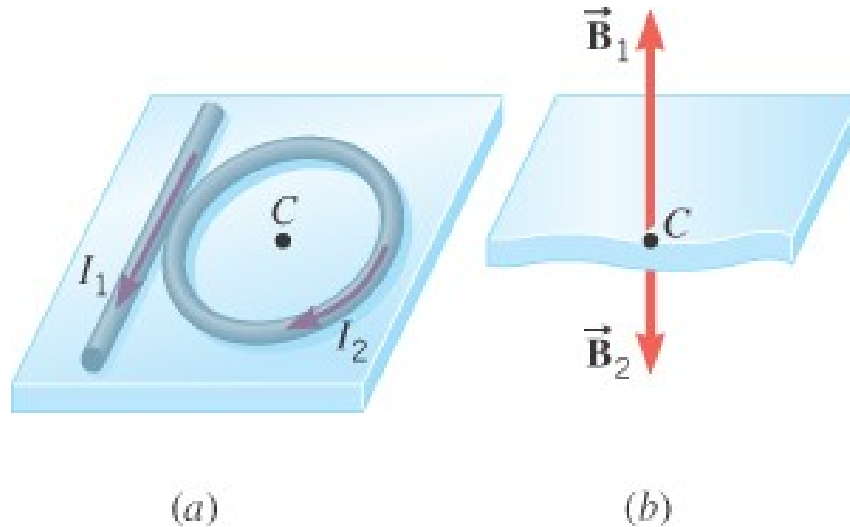
(a)



(b)

21.7 Magnetic Fields Produced by Currents

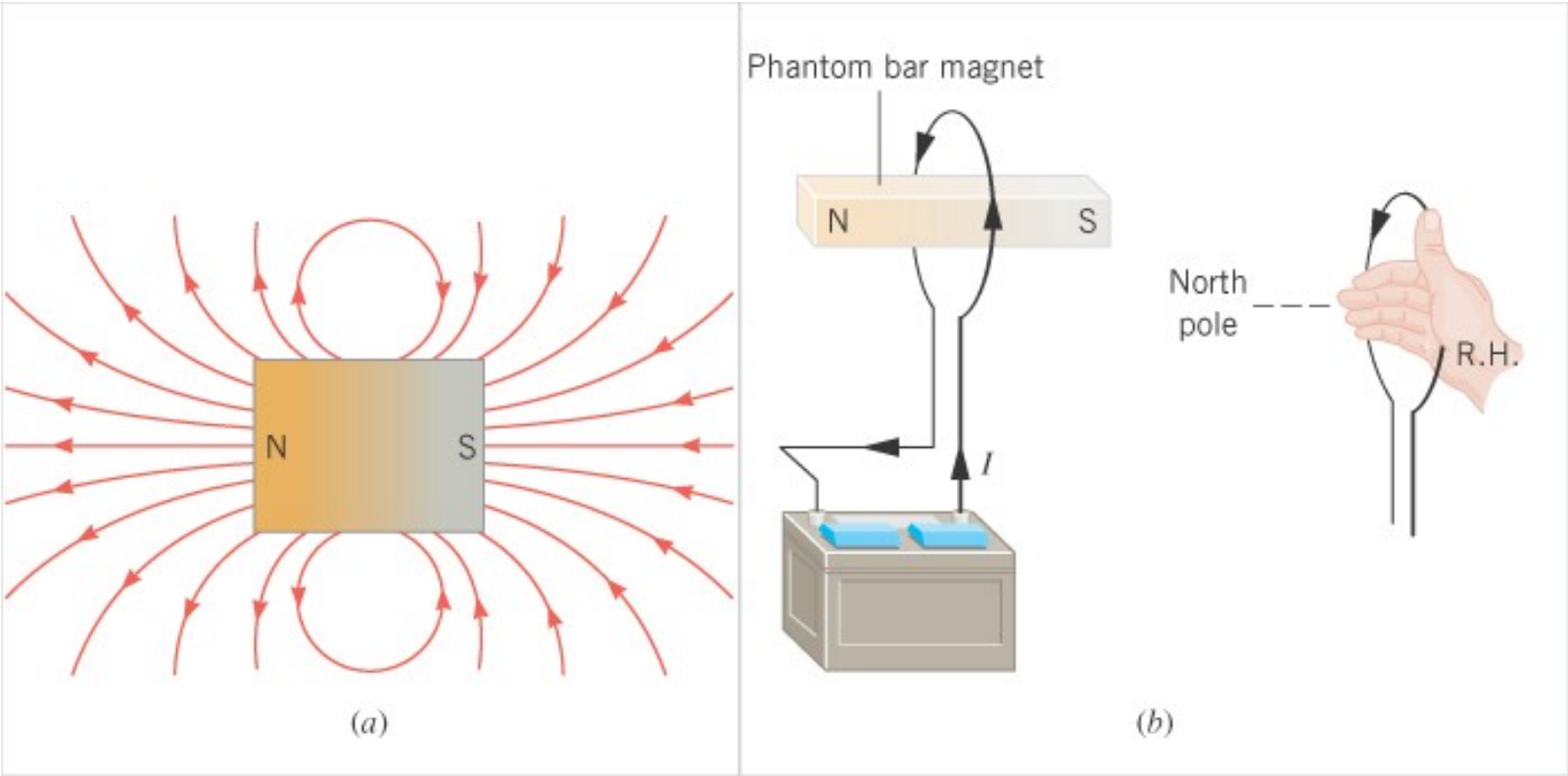
$$B = \frac{\mu_o I_1}{2\pi r} - \frac{\mu_o I_2}{2R} = \frac{\mu_o}{2} \left(\frac{I_1}{\pi r} - \frac{I_2}{R} \right)$$



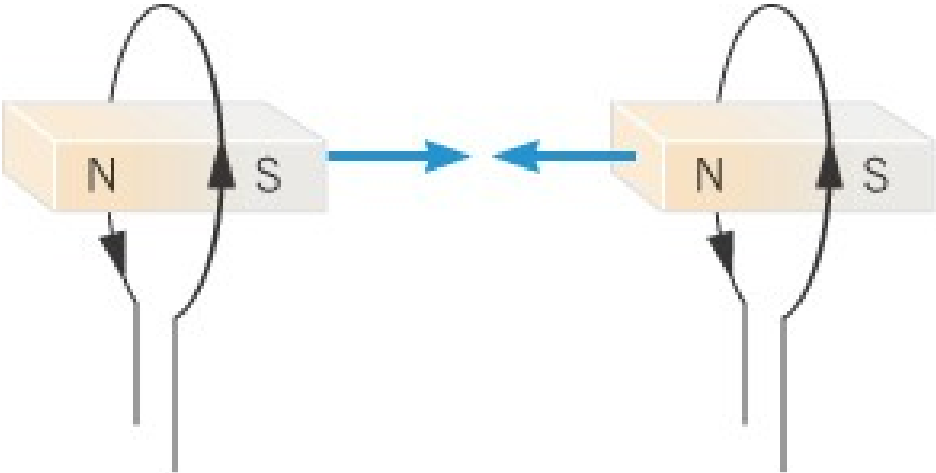
$$B = \frac{(4\pi \times 10^{-7} \text{ T}\cdot\text{m/A})}{2} \left(\frac{8.0 \text{ A}}{\pi(0.030 \text{ m})} - \frac{2.0 \text{ A}}{0.030 \text{ m}} \right) = 1.1 \times 10^{-5} \text{ T}$$

21.7 *Magnetic Fields Produced by Currents*

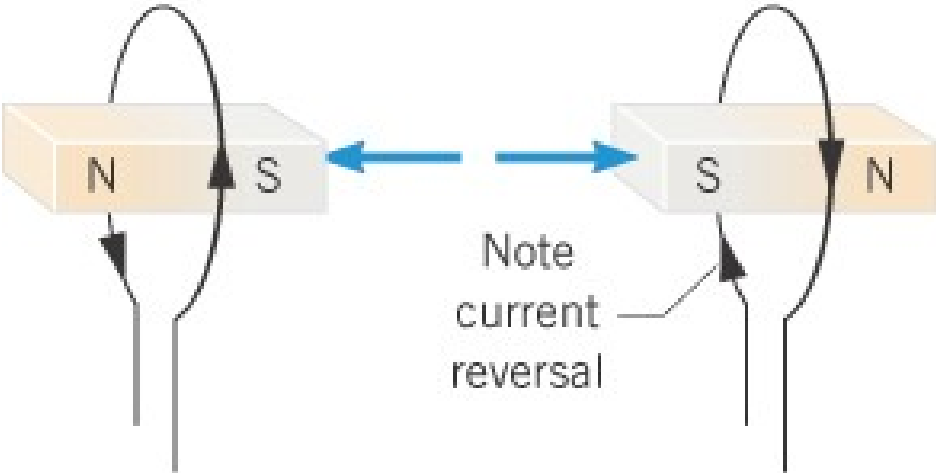
The field lines around the bar magnet resemble those around the loop.



21.7 Magnetic Fields Produced by Currents



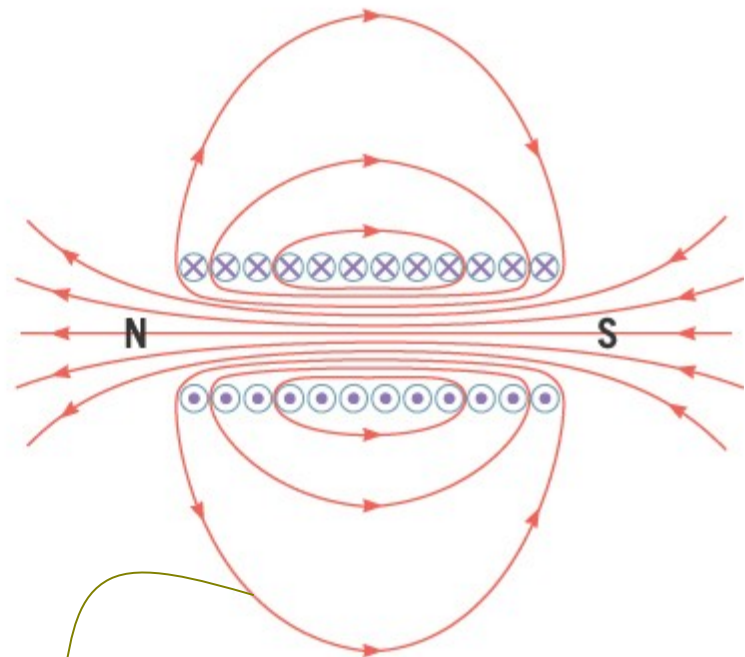
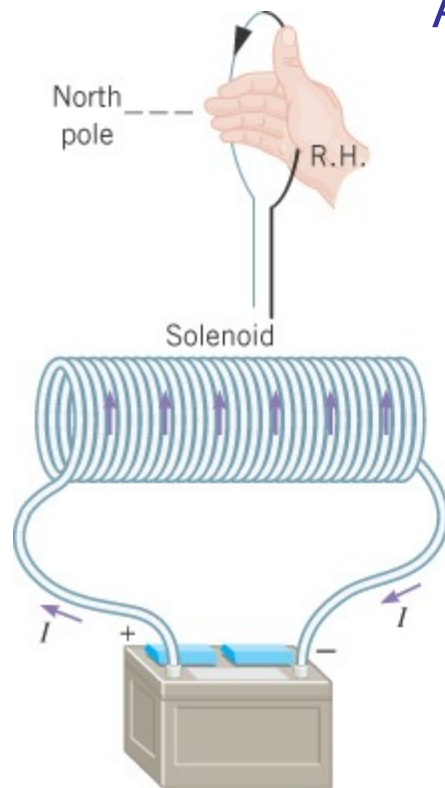
(a) Attraction



(b) Repulsion

21.7 Magnetic Fields Produced by Currents

A SOLENOID



Interior of a solenoid

$$B = \mu_0 n I$$

number of turns per
unit length

If the number of turns is x 5 then B is ?

2 layers of wire, with 40 turns each, are wound onto a coil from 15cm long and a Current of 0.30A is passed through the wire. What is the magnetic field inside the solenoid

21.7 Magnetic Fields Produced by Currents An electric current produces a magnetic field, with different current geometries giving rise to different field patterns. For an infinitely long, straight wire, the magnetic field lines are circles centered on the wire, and their direction is given by Right-Hand Rule No. 2 (see below). The magnitude of the magnetic field at a radial distance r from the wire is given by Equation 21.5, where I is the current in the wire and μ_0 is a constant known as the permeability of free space ($\mu_0 = 4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}$).

$$B = \frac{\mu_0 I}{2\pi r} \quad (21.5)$$

Right-Hand Rule No. 2: Curl the fingers of the right hand into the shape of a half-circle. Point the thumb in the direction of the conventional current I , and the tips of the fingers will point in the direction of the magnetic field $\vec{B}$.

The magnitude of the magnetic field at the center of a flat circular loop consisting of N turns, each of radius R and carrying a current I , is given by Equation 21.6. The loop has associated with it a north pole on one side and a south pole on the other side. The side of the loop that behaves like a north pole can be predicted by using Right-Hand Rule No. 2.

$$B = N \frac{\mu_0 I}{2R} \quad (21.6)$$

A solenoid is a coil of wire wound in the shape of a helix. Inside a long solenoid the magnetic field is nearly constant and has a magnitude that is given by Equation 21.7, where n is the number of turns per unit length of the solenoid and I is the current in the wire. One end of the solenoid behaves like a north pole, and the other end like a south pole. The end that is the north pole can be predicted by using Right-Hand Rule No. 2.

$$B = \mu_0 n I \quad (21.7)$$

21.8 Ampère's Law Ampère's law specifies the relationship between a current and its associated magnetic field. For any current geometry that produces a magnetic field that does not change in time, Ampère's law is given by Equation 21.8, where $\Delta\ell$ is a small segment of length along a closed path of arbitrary shape around the current, $B_{\parallel}$ is the component of the magnetic field parallel to $\Delta\ell$, I is the net current passing through the surface bounded by the path, and μ_0 is the permeability of free space. The symbol Σ indicates that the sum of all $B_{\parallel} \Delta\ell$ terms must be taken around the closed path.

$$\Sigma B_{\parallel} \Delta\ell = \mu_0 I \quad (21.8)$$

21.9 Magnetic Materials Ferromagnetic materials, such as iron, are made up of tiny regions called magnetic domains, each of which behaves as a small magnet. In an unmagnetized ferromagnetic material, the domains are randomly aligned. In a permanent magnet, many of the domains are aligned, and a high degree of magnetism results. An unmagnetized ferromagnetic material can be induced into becoming magnetized by placing it in an external magnetic field.